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RESEARCH

Digital twin-enabled deep learning for real-time fire situation awareness

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Abstract

The complexity of a building's indoor environment exacerbates its fire safety issues, making them particularly problematic. Traditional fire protection systems only detect fires when their characteristics are already visible, and fire recognition delays have become a significant problem. This paper presents a digital twin framework for managing building fire safety in real time. Sensor data and a Bert_C model are used to analyze the transition of a fire from smoldering to flaming combustion. The results show that the model achieved 96.1% accuracy and 96.2% F1 score, and the framework significantly improves the timeliness and accuracy of fire warnings and visualizes 3D fire scenarios, providing powerful technical support for firefighting and emergency rescue operations. This study confirms the feasibility of real-time fire safety management using 3D environments and digital twins, providing an innovative solution for fire safety management in the built environment.

Keywords Digital twin · Deep learning · Building fire · AIoT · Numerical simulation

1 Introduction

Amidst rapid economic development and urbanization, the density of urban buildings has grown, significantly boosting the efficiency of how residential and commercial spaces are used. While these structures offer considerable convenience, they also present significant fire risks. Building fires, due to their distinct physical settings and architectural traits, often lead to serious casualties and considerable economic damage, affecting both the occupants and the firefighting personnel. Data from the Fire and Rescue Bureau of the Ministry of Emergency Management[1] reveal that from 2012 to 2021, there were 1,324,000 residential fires across the country, resulting in 11,634 deaths, 6,738 injuries, and direct property losses totaling 7.77 billion yuan. Among these incidents, fires in university laboratories are particularly common and dangerous. For example, the 2021 laboratory fire at Nanjing University of Aeronautics and Astronautics and the 2019 incident at Nanjing University of Technology both caused significant casualties and garnered widespread public attention. These events underscore the pressing need to improve fire safety measures in university laboratories.

Various studies have aimed to decrease the likelihood of fires by implementing stringent fire codes and by developing automated systems for the early detection and suppression of fires [1, 2]. Additionally, some research has concentrated on bolstering the fire resistance of buildings through the use of fire-resistant coatings [3] and by refining structural designs [4, 5]. While these measures do reduce fire risks to a degree, they are not fully effective in preventing uncontrollable fire incidents. Unforeseen elements such as human errors and surrounding factors can still lead to devastating fires that result in considerable loss of life and property.

In recent years, the swift progress in Internet of Things (IoT) and machine learning (ML) technologies has made smart firefighting a leading research field. These advancements not only improve early fire detection but also enhance emergency responses and post-disaster recovery in urban

[1] Source:CCTV New Client 2022–2-18. http://www.cneb.gov.cn/2022/02/18/ARTI1645166137776548.shtml

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areas affected by fires [6]. Smart firefighting systems facilitate automated escape route planning and expedite the real-time transmission of fire scene data, offering decision-makers more precise and timely support [7]. In the realm of intelligent decision-making for fire rescue, Jaber et al. [8] introduced TeleGeoProcessing and intelligent software in 2001, which have been successfully piloted in forest firefighting. This demonstrates the substantial potential of computer technology in aiding complex fire rescue operations. Following these developments, Zhang et al. [9] created a digital twin system for real-time fire safety management in tunnels. This system combines dynamic sensor data with AIoT technologies to predict fire locations and sizes in real time. Moreover, the digital twin platform visualizes 3D fire scenes to support evacuation, firefighting, and emergency rescue, providing an efficient decision-support system for firefighting operations.

The practical application of AI models in firefighting relies on well-developed systems for data communication, visualization, and human–computer interaction. In this context, digital twin technology has become a key component of smart buildings and firefighting systems [10]. Liu et al. [11] proposed a digital twin-driven dynamic guidance method for firefighting evacuation that integrates building information modeling (BIM) and IoT technologies. This method constructs an evacuation-focused digital twin model capable of real-time surrounding data collection, 3D visualization of indoor layouts, fire alarm activation, indoor personnel localization, and evacuation path planning. Furthermore, Jiang et al. [12] introduced the integration of digital twin and semantic web technologies for intelligent control of building fire protection systems in various fire scenarios. This approach enables intelligent monitoring and regulation of fire protection systems, allowing them to respond more adaptively and effectively to diverse fire conditions.

This study proposes a digital twin system that integrates IoT and AIoT technologies, aiming to comprehensively optimize building fire safety management systems to minimize casualties and property losses from fires. The system uses deep learning algorithms and numerical simulation techniques for rapid identification and quantitative analysis of fire incidents. With digital twin technology, the system extends fire management capabilities by incorporating advanced visualization tools, significantly enhancing emergency response effectiveness. The system also integrates a fire-monitoring digital twin platform developed with the Unity 3D game engine, offering administrators a real-time visualization interface for fire scenes, thereby greatly improving the system's practical applicability. Through this interface, administrators can monitor fire progression, assess associated risks accurately, and develop scientifically grounded emergency plans. In the event of fire emergencies, the system ensures swift responses by providing real-time, comprehensive decision-making support to administrators, facilitating the effective implementation of emergency measures.

2 Related work

2.1 Digital twin

The concept of the digital twin was first introduced by Dr. Grieves in 2003 under the term "Mirrored Space Model" [13]. The initial framework of the digital twin was identified as comprising physical products, virtual products, and their interconnections [14]. Gabor et al. [15] described the digital twin as an advanced simulation created using domain expertise and real-world data, capable of achieving accurate simulations across varying temporal and spatial scales. Tao et al. [16] emphasized that digital twin modeling serves as the core of representing physical entities and, driven by data and models, enables monitoring, simulation, and prediction. Subsequently, Tao and Zhang [17] proposed that a comprehensive digital twin comprises five key components: physical entities, virtual entities, interconnections, data, and services, with data being fundamental for generating new knowledge. These elements collectively form the core structure of the digital twin [18]. A digital twin is not merely a digital representation of a physical entity; it also facilitates real-time acquisition and analysis of operational, surrounding, and status-related information, thereby supporting decision-making processes. With advancements in the Internet of Things (IoT), big data, artificial intelligence, and high-performance computing, the concept of the digital twin has evolved and found applications beyond manufacturing. Its adoption has expanded into diverse fields, including smart cities, healthcare, aerospace, and transportation.

Digital twin technology represents a relatively novel approach in the domain of building fire safety. To address the increasingly stringent requirements for fire safety in complex and dynamic building environments, a digital twin framework integrating IoT, building information modeling (BIM), and advanced machine learning techniques has been developed. This framework enables real-time monitoring and dynamic simulation of critical elements, including the internal environment, fire protection systems, and personnel distribution within a building, by constructing a digital representation of the physical structure. These findings underscore the framework's potential to provide robust technical support and a reliable decision-making foundation for intelligent firefighting applications.

2.2 Intelligent fire fighting

In recent years, research on the application of artificial intelligence (AI), virtual reality (VR), IoT, and BIM in conjunction

with digital twins (DT) for building fire protection has gained significant attention. Artificial intelligence models trained on extensive datasets—including simulation, experimental, and historical data—have demonstrated the capability to predict real-time fire-related information, such as fire size and location. Furthermore, such models can forecast critical events, including flashover, backdraft, or structural responses during fire incidents. The integration of deep learning algorithms with flame imagery has shown great potential for fire recognition. For instance, Mao et al. [19] proposed an innovative fire recognition approach leveraging multi-channel convolutional neural networks. This method applies to a wide range of fire scenarios and types, achieving an impressive classification accuracy of 98%. In addition to fire recognition, real-time prediction of the heat release rate (HRR) of fires is now feasible. Zhang et al. [20] developed an intelligent model utilizing AI algorithms to predict real-time fire information and critical events. This approach efficiently identifies dynamic flame behavior and accurately determines transient fire heat release rates. Zhang et al. [21] proposed an innovative artificial intelligence-digital-fire (AID-Fire) framework, designed to facilitate real-time recognition of complex fire scenarios in buildings. Experimental results demonstrate that the AID-Fire system achieves efficient fire information recognition, with a relative error within 15%.

2.2.1 BIM application

The application of BIM in fire safety management has been extensively studied [22], highlighting its significant potential in enhancing fire prevention and response systems. BIM not only facilitates the digital creation and management of building models but also serves as a foundational tool for constructing digital twins. For instance, Cheng et al. [23] developed a BIM-based intelligent fire rescue system that integrates sensor technologies, such as Bluetooth sensors, to enable real-time monitoring of fire-related information. This system efficiently displays fire data on mobile devices, significantly improving early fire detection and expediting rescue operations. Moreover, BIM has been applied to emergency evacuation planning.

The application of building information modeling (BIM) technology in building fire protection is primarily demonstrated in its role in intelligent decision-making. For instance, Cheng et al. [23] developed a BIM-based intelligent fire prevention and disaster relief system, which implements personal localization, evacuation/rescue route optimization using Bluetooth technology, and mobile guidance. This system also enables 3D visualization to support fire safety assessment and planning.

Although building information modeling (BIM) has been widely adopted in the fields of architecture and construction, its current applications remain largely confined to static representations of physical assets. This inherent limitation restricts BIM's capacity to dynamically integrate with real-time data streams generated by fire safety sensors and Internet of Things (IoT) devices. Conventional BIM models typically serve as static digital twins of the built environment and lack the interoperability required to ingest and process time-sensitive environmental data—such as temperature variations or smoke detection alerts—produced by IoT infrastructures like LoRaWAN or NB-IoT. Consequently, this disconnect hinders the establishment of real-time interactions between BIM and fire safety systems, thereby reducing the operational effectiveness of emergency response mechanisms in smart building contexts.

Teizer et al. [24] emphasize the potential of integrating BIM with IoT sensor networks to address this shortcoming. By aligning the spatial representation capabilities of BIM with the temporal resolution of IoT-derived data, such integration facilitates proactive facility management through continuous monitoring of key fire safety indicators. For instance, sensor-triggered events—such as abnormal increases in temperature—can be visualized in real time within the BIM environment. This enables facility managers to pinpoint high-risk zones, optimize evacuation planning, and coordinate emergency interventions with enhanced spatial accuracy and operational precision.

2.2.2 Bert model

Deep learning-based text categorization models cover a wide range of architectures such as CNN models, RNN models, and Bert models. In the early days of the natural language field, the TextCNN text classification model came to the forefront, which simplified the convolutional layers based on the original network structure and improved the training speed, but the model interpretability was lacking. Lai et al. [25] proposed a recurrent convolutional neural network for text classification without human-designed features. To solve the short-term memory of RNN, LSTM and GRU were proposed one after another, and then the Bert model was proposed as a pre-training model, which does not need a large amount of corpus for training in a specific scenario, and it has the advantages of strong generalization ability and simplicity of use. Chen et al. [26] proposed a Bert-based local feature convolutional network (LFCN) model to enhance the accuracy of long-text classification in Chinese news.

Shi el al. [27] proposed a BERT-CNN network model based on the combination of pre-trained BERT network and convolutional neural. The BERT structure is used to express sentence semantics as text vectors, and then the local features of the sentences are extracted by convolutional neural network, and the experiments on the labeled dataset of reviews of a cell phone in Jingdong show that the method has good performance in this field. Therefore, this paper proposes to

combine Bert model and CNN model together, which not only can solve the problem of insufficient corpus, but also can play the ability of CNN to extract local features.

2.2.3 Sensor data

Due to the inherent challenges associated with conducting large-scale fire tests and generating extensive experimental datasets, the integration of machine learning (ML) with numerical fire simulations has been proposed as a viable approach for training predictive models. For instance, Wang et al. [28] developed a prediction model known as P-Flash, which effectively forecasted the risk of flashover in multi-room compartment fires. This model was trained using heat detector temperature data obtained from numerical simulations. Furthermore, machine learning techniques have also been employed to predict smoke movement during tunnel fires.

Data extracted from fire scenes can be used to predict fire occurrence and enhance the effectiveness of fire monitoring. To address the challenges in fire detection, Li et al. [29] developed an integrated dual-parameter fire detector based on mid-infrared technology, which simultaneously measures carbon monoxide (CO) and carbon dioxide (CO_2) concentrations. This innovation was further complemented by a mobile fire detection method. Similarly, Solórzano et al. [30] proposed utilizing an array of gas sensors to monitor CO concentrations at fire scenes, enabling the generation of early warning information. Schmitt et al. [31] employed gas detection techniques for CO, O_2, and other gases to facilitate fire detection and identification, further enhancing the accuracy of fire warning systems. Additionally, Deng et al. [32] demonstrated that the light scattering intensity of fire smoke is correlated with the volume and surface area of smoke particles.

To address the challenges associated with intelligent decision-making, and sensor data, this study proposes a digital twin framework that integrates sensor data with an enhanced Bert model. The framework is designed to simulate and establish dynamic connections between real-world and virtual buildings.

3 Proposed digital twin framework

Digital twins are virtual representations of responsive physical entities. According to the white paper by Gerives, digital twins comprise three key components: (1) physical entities in the real world, (2) a virtual model in the digital domain, and (3) the interaction between physical entities and their digital representations. Figure 1 illustrates the general framework of an architectural digital twin.

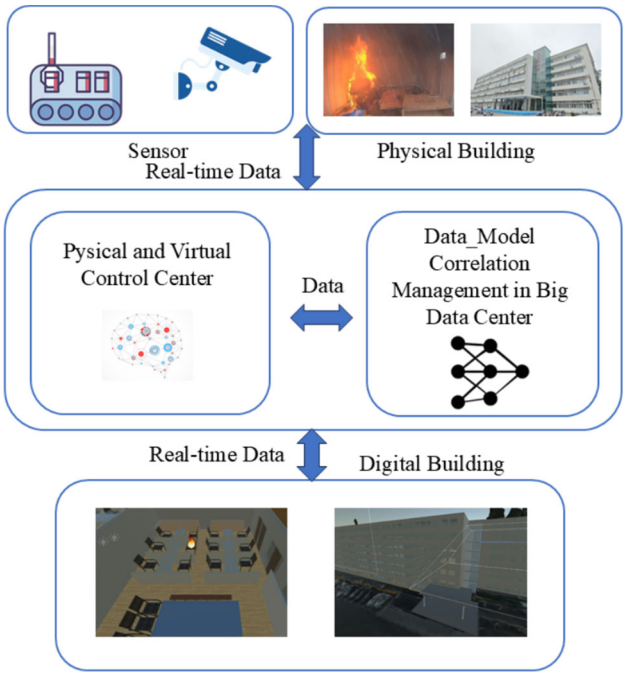


Fig.1 The general framework of the architectural digital twin

The physical entity of the digital twin model corresponds to the real-world building, while the virtual model represents the digital building. The exchange of data between the real building and its digital counterpart establishes a connection between the physical and virtual spaces. The digital twin model not only facilitates real-time recognition and prediction of building fires but also supports emergency fire safety response. Furthermore, it can be integrated with other fire protection systems to enhance operational control and emergency functionality. Additionally, the digital twin model provides a three-dimensional visualization platform that enables real-time representation of fire scene conditions. Through this platform, users can intuitively interact with physical entities using a virtual interface.

3.1 Interactivate operations

The current digital twin framework primarily collects data through sensors and CCTV cameras to enable the dynamic detection and visualization of fire scenarios. Figure 2 illustrates the management process of the indoor fire digital twin system, which can be categorized into three main phases: data collection, control platform, and data visualization.

In the data collection phase, the system employs a variety of sensors to systematically gather data. These sensors capture critical values, including smoke concentration, temperature, and carbon monoxide levels, which serve as fundamental inputs for fire warning systems and situational analysis.

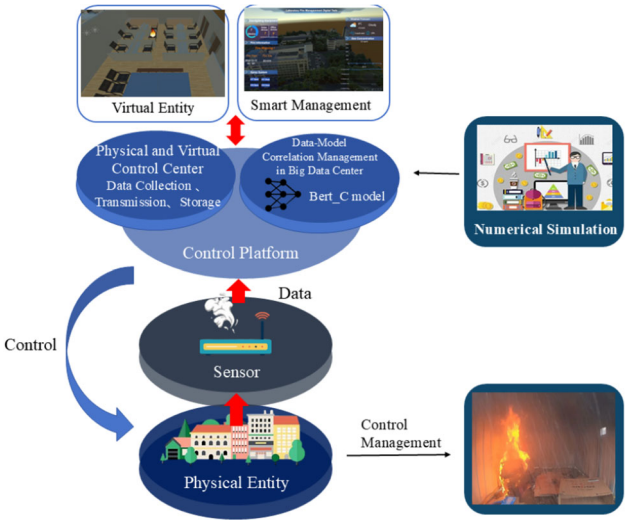


Fig.2 Management flow of the digital twin system: Control platform contains data storage, data transmission, and data processing stages

In the data transmission phase, the fire safety digital twin system can utilize either wired or wireless communication methods. Wired transmission offers stability, high bandwidth, and strong resistance to interference; however, it involves complex wiring and incurs high maintenance costs. In contrast, wireless transmission is characterized by flexibility, low costs, and adaptability but is more vulnerable to electromagnetic interference and signal obstruction. To enhance the overall performance of the system, a hybrid transmission strategy is proposed. In this approach, critical data are transmitted via wired channels, while non-critical data are conveyed through wireless channels. This strategy effectively improves both transmission efficiency and the system's resistance to interference, ensuring the reliable operation and precise detection capabilities of the fire safety digital twin system.

In the data processing phase, the system undertakes data cleaning, analysis, and fusion of raw sensor data to enable fire status prediction. Data cleaning serves as a fundamental step to ensure data quality, wherein preprocessing techniques are applied to remove noise and inconsistencies. Following this, the cleaned data are subjected to advanced analytical methods. Specifically, deep learning models are employed to conduct an in-depth analysis of the sensor data. These models identify patterns indicative of fire occurrences, thereby facilitating the prediction of potential fire scenarios. Furthermore, the system employs a multi-feature and multi-source data fusion strategy to integrate information from various devices, including smoke sensors, temperature and humidity sensors, and CCTV cameras. This integration significantly enhances the accuracy of fire predictions by synthesizing diverse data inputs, thereby providing a more comprehensive and reliable analysis of fire dynamics.

In the data visualization phase, the system conveys the processed data through various visual representations, including line graphs and real-time monitoring charts. These visual tools enable administrators to monitor dynamic trends in sensor data in real-time, facilitating the identification of outliers or abnormal variations. Consequently, potential fire hazards can be promptly targeted and addressed.

In addition, the system incorporates an interactive user interface that facilitates bidirectional communication between the equipment and the user. Leveraging advanced 3D modeling and virtual reality technologies, the system constructs dynamic fire scenarios, enabling administrators to visualize real-time fire data and the status of the fire scene. Furthermore, the system allows for remote control of indoor firefighting equipment, including sprinkler systems and emergency alarms, enhancing its utility in fire management and response.

3.2 Sensor data

In the event of a building fire, CCTV camera systems can monitor the fire scene; however, their coverage is often limited due to cost constraints and privacy protection concerns. Furthermore, in actual fire scenarios, the presence of smoke frequently obstructs the camera's field of view, rendering continuous monitoring ineffective. To address these challenges more effectively, fire events can be identified by analyzing gas concentrations. This study employs a sensor system that integrates smoke, temperature, and humidity detection to accurately quantify smoke concentration and surrounding environmental conditions. Additionally, a portable smoke analyzer equipped with built-in CO and O_2 sensors is utilized to monitor changes in CO and O_2 concentrations during fire incidents.

Pfister et al. [33], during a series of experiments aimed at monitoring and recording the progression of several standardized fire scenarios, observed that the concentration of carbon monoxide (CO) increased notably after a certain period of fire development, reaching levels significantly higher than the ambient CO concentration in non-fire conditions. This observation underscores the potential of CO concentration changes as a critical indicator of fire onset, thereby offering substantial value for early fire detection and warning systems. Similarly, Du et al. [34] demonstrated that considerable amounts of carbon monoxide (CO) and carbon dioxide (CO_2) are generated during the initial phase of combustion. These findings reinforce the notion that the variations in CO and CO_2 concentrations are particularly prominent in the early stages of a fire, making them reliable indicators for timely detection. Moreover, as evidenced by the findings of Study Cleary et al. [35] integrating smoke concentration measurements with CO concentration monitoring reduces the false alarm rate by approximately 40% compared

to the use of smoke detection alone. This significant improvement highlights the essential role of smoke concentration as a core parameter in fire detection systems.

Accordingly, this study selects smoke concentration, CO concentration, CO_2 concentration, and temperature as the primary detection parameters to provide robust support for the early identification of fire incidents.

In this study, AUTBUS bus technology is utilized for both sensor data acquisition and device control. The system supports a bandwidth of up to 100 Mbps and allows the integration of up to 254 nodes, thereby fulfilling the real-time requirements for multi-sensor data collection. Moreover, AUTBUS supports a communication distance of up to 500 m, making it suitable for long-range transmission in large or complex environments. A bus-type network topology is adopted, in which all relevant components—including sensors, sprinkler heads, and fire alarms—are connected to the AUTBUS network. This configuration establishes a unified and scalable communication framework. Within this architecture, AUTBUS facilitates not only the real-time transmission of environmental parameters, such as smoke concentration, temperature, and gas levels, to the central processing unit, but also the dispatch of control signals to corresponding actuators. For instance, it enables the activation of sprinklers or alarm modules upon the detection of fire-related risks.

In order to ensure the accuracy of the sensor data, this paper transmits the data with the system through the RS485 communication protocol, which has been certified by the IEC 61158–2 standard. This protocol has a signaling delay of about 5 ns/m, which provides high timeliness and reliability in fire-monitoring applications.

3.3 Digital twins

The digital twin system presented in this study is specifically designed to enhance indoor fire safety management by providing efficient fire facility management and control, fire information visualization, fire risk assessment, and evacuation and rescue guidance capabilities. By integrating multiple advanced technologies, the system establishes a real-time, data-driven platform that seamlessly fuses virtual and physical environments, offering intelligent management solutions for indoor fire safety.

Regarding fire facility management and control, the system enables real-time monitoring of the operational status of fire safety equipment through the digital twin model, including systems such as water sprinklers, smoke alarms, and other related devices. Furthermore, the system incorporates a remote-control function, allowing managers to directly operate the equipment via the interface, such as activating the sprinkler system or deactivating air conditioning and other devices, thereby ensuring prompt response in case of an emergency.

Regarding fire information visualization, the system integrates various data collected by sensors, such as smoke concentration and carbon monoxide levels, and presents the indoor fire situation in real time through dynamic curves and 3D modeling. Additionally, by incorporating camera data, the system enables visualization of the fire scene, offering managers a comprehensive basis for assessing the situation and ensuring the accuracy and timeliness of decision-making.

Regarding fire risk assessment, the system employs deep learning models to analyze and predict the real-time data collected by the sensors. By identifying whether a fire has reached the critical point of transitioning from smoldering to flaming combustion, the system can implement suppression measures in advance, thereby effectively preventing the further spread of the fire and mitigating the risk of property damage and casualties.

Regarding evacuation and rescue guidance, the system offers intelligent support for evacuation and rescue operations during emergencies, such as fires. The system's evacuation indicators (e.g., escape route lights and fire alarm devices) provide escape guidance for individuals who are trapped. Additionally, the system supplies rescuers with detailed disaster data and surroundings information, including the site of the fire origin and the distribution of high-temperature zones, thereby ensuring the efficiency and safety of rescue operations.

4 Implementation of the digital twin framework

The real-time performance of fire recognition models is critically influenced by the availability of high-quality datasets. However, conducting fire experiments in real-world environments presents significant challenges, as such experiments are both costly and complex. Consequently, numerical simulation has emerged as an effective alternative. In this study, numerical simulations were performed to assess the performance of the fire recognition model.

4.1 Numerical simulation

FDS is a fire simulation software based on field simulation developed by the Building and Fire Research Laboratory of the National Institute of Standards and Technology. The software adopts advanced vortex simulation technology, which has been verified by many examples and is widely used in the field of fire safety engineering. Field simulation, also known as physical simulation, is a high-level complex simulation based on the basic equations of mass, momentum, energy, and chemical reaction of the fire process. The simulation

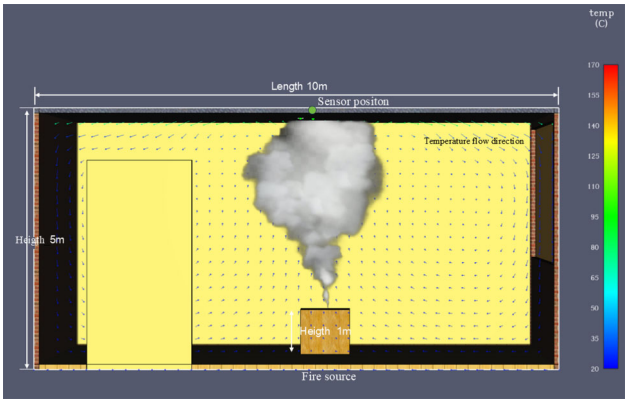


Fig.3 Fire room in FDS simulation:

can be solved to obtain a series of data such as temperature, CO concentration, CO2 concentration, O2 concentration, and visibility at the relevant measurement points [36].

FDS software is the most widely used software in the field of fire simulation. From the engineering applications and scientific and technological literature, 90% or even more of the fire simulations are completed by FDS, which is determined by the inherent characteristics of FDS. This software fire simulation is feature-rich, post-processing powerful and open source free [37, 38].

Fire simulation has traditionally been conducted using computational fluid dynamics (CFD)-based software, including FDS, CFAST, and other related tools [39, 40]. Hodges et al. [41] demonstrated that CFD techniques can effectively replicate fire scenarios, facilitating the generation of datasets for training fire prediction models. Gao [42] used PyroSim to model and perform FDS combustion simulations, confirming that it is feasible to use FDS for numerical simulation of fires in conventional dwellings.

In this study, the 3D design software PyroSim was employed as the simulation platform for fire dynamics analysis. Using a laboratory in the information science building of a university as a prototype, a virtual 3D laboratory model was constructed at a 1:1 scale. To simplify the model, certain non-essential components were omitted during the design process. The geometric dimensions of the laboratory, measuring 10.0 m × 6.0 m × 5 m (length × width × height), were meticulously replicated to reflect the actual spatial layout, thereby ensuring the accurate simulation of fire scenarios. The height of the burning column is 1 m. Ventilation conditions were excluded from consideration in the model. Since paper is widely utilized in laboratories as a primary medium for storing documents, books, and experimental records, it was selected as the principal combustion material. In addition, in this paper, the initial value of heat of combustion of wood is set as 15 kW/m2. The calculation model diagram of FDS is shown in Fig. 3.

During the modeling process, the laboratory door was configured to remain closed, and various sensors, including smoke sensors and temperature sensors, were strategically arranged on the ceiling to facilitate real-time data collection. All fire simulations conducted in this study utilized a grid size of 0.15 m, with a total simulation duration of 100 s. Following the specification of the time step, the configuration of numerical algorithms, and the definition of output parameters, the PyroSim software was executed to simulate fire dynamics. Output parameters, such as smoke particle concentration, CO concentration, and O_2 concentration, were continuously monitored within the model.

To eliminate potential influences of the simulation results are not affected by the grid resolution, grid independence tests were conducted in this paper. The predetermined convergence criteria for the parameters are smoke concentration < 5%, CO concentration < 8%, CO_2 concentration < 5%, and temperature < 3%. In this paper, a monitoring point was arranged at 1.5 m above the fire source using smoke concentration, CO concentration, CO_2 concentration, and temperature as parameters. The grid sizes used in this paper are coarse grid size 0.25 m × 0.25 m × 0.25 m, medium grid size 0.15 m × 0.15 m × 0.15 m, and fine grid size 0.12 m × 0.12 m × 0.12 m. In this paper, error analysis is performed using a comparison of the medium and fine grids. The error analysis is quantified using well-defined error formulas, see Eq. 1.

$$\text{Error} = \left| \frac{\text{Ømedium} - \text{Øfine}}{\text{Ømedium}} \right| \times 100\% \tag{1}$$

In the calculation results of the medium grid, the error analysis was performed for the key parameters of smoke concentration, CO concentration, CO_2 concentration, and temperature. After independent testing, the smoke concentration is 15.2 mg/m^3, 18.6 mg/m^3, 19.1 mg/m in the coarse, meso, and fine grids, respectively, with an error of 2.6%, which is lower than the preset criterion of 3%; the CO concentration is 450 ppm, 522 ppm, 550 ppm in the coarse, meso, and fine grids, respectively, with an error of 5.1%, which is lower than the preset criterion of 8%; CO2 concentration is 1.2%, 1.53%, 1.6% in the coarse, medium, and fine grids, respectively, and its error is 4.3%, which is lower than the preset standard of 5%; and temperature is 280 °C, 313 °C, 320 °C in the coarse grid, respectively, and its error is 2.2%, which is lower than the preset standard of 3%.

Based on the above error analysis results, it can be determined that the smoke concentration, CO concentration, CO2 concentration, and temperature show good convergence under the medium grid.

When a fire transitions from the smoldering to flaming combustion, the rate of pyrolysis and the combustion of flammable materials increase significantly, leading to notable

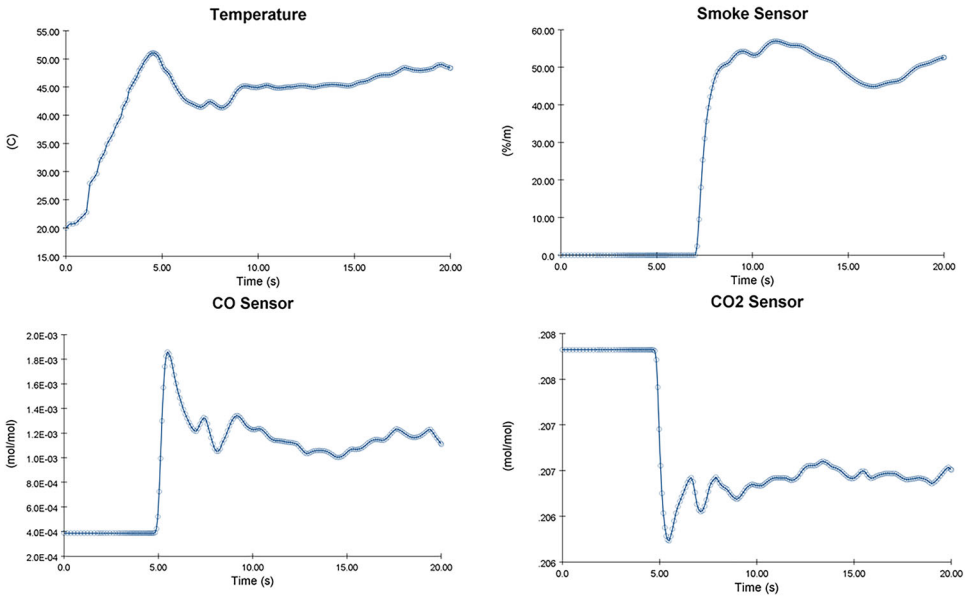


Fig.4 Indoor FDS gas concentration distribution: **a** The upper left panel is indoor temperature data, **b** the upper right panel is indoor smoke concentration data, **c** the lower-left panel is indoor CO concentration data, **d** the lower right panel is CO2 concentration data

changes in fire dynamics. At this critical juncture, the concentration of smoke rises rapidly, and the concentration of CO increases markedly, while the concentration of O_2 gradually decreases. Figure 4 illustrates the dynamic variation curves of smoke concentration, CO concentration, CO_2 concentration, and temperature during the combustion process, which provide a robust scientific basis for analyzing changes in fire behavior and their environmental impact.

In the context of data processing, this study employs numerical simulation to construct a dataset comprising 1210 entries, primarily focusing on parameters such as smoke concentration, CO concentration, CO_2 concentration, and temperature generated during the fire scenario. Subsequently, 100 additional data points were acquired through fire simulation in a realistic scenario for validation purposes. The acquired dataset underwent data preprocessing, where the experimental data were consolidated into a standardized format. Finally, the dataset was divided into a training set and a validation set, with a ratio of 7:3.

To further investigate the dynamics of fire transition and enhance the accuracy of fire information prediction, this study generates a dataset of fire transition from smoldering to flaming combustion through numerical simulation. By integrating key concentration change data, collected in real time by fire sensors, a deep learning model based on the BERT architecture is developed. This model leverages the temporal characteristics of the data, enabling intelligent prediction of fire-related information. The resulting predictions offer valuable support for fire situational awareness and emergency response efforts.

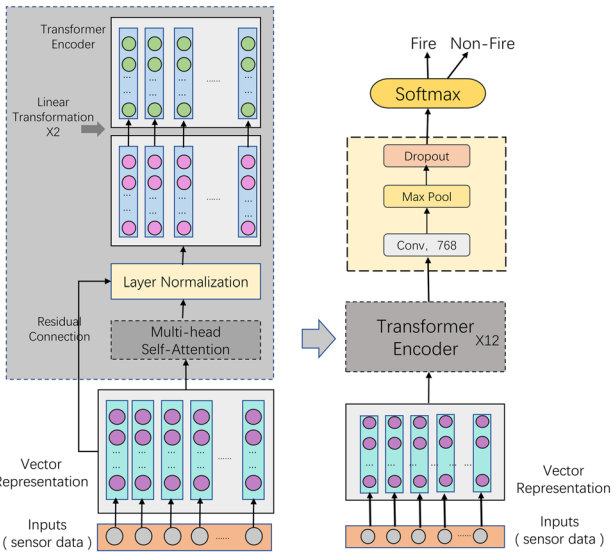


Fig.5 Architecture of the proposed Bert model

4.2 Improved bert structure

To address the issue of early-stage classification of both smoldering and flaming combustion scenarios, this paper introduces the Bert_C model, the architecture of which is depicted in Fig. 5. The Bert model has demonstrated strong performance in short-term time series classification tasks, while convolutional neural networks (CNN) are particularly effective at capturing local features. To enable the model to capture both local and global features, this paper integrates the Bert and CNN models. Initially, the bidirectional transformer feature extractor of the Bert model is used to extract contextual features from the time series data. Subsequently,

the CNN is responsible for identifying local features and pattern changes within a specified time window. After the text vectors reach the Bert layer, an attention mechanism is utilized to extract global features. In the CNN layer, convolutional operations are employed to extract locally significant features, such as key phrases within the text. A max-pooling layer is also employed, which automatically identifies the words that play a critical role in text classification, thereby capturing the key components within the text. The feature vectors generated from these two layers are then fused, and the predictions are converted to probability values as the results using the softmax function.

In this study, data on smoke concentration, carbon monoxide (CO), carbon dioxide (CO_2), and temperature, obtained from multiple types of sensors, are utilized as input features for an enhanced BERT-based model. These inputs are first encoded into vector representations and subsequently passed through a 12-layer transformer encoder. Notably, only the outputs from layers 9 through 12 are employed in this work, as each encoder layer incorporates a multi-head self-attention mechanism, residual connections, and layer normalization, enabling the model to capture complex dependencies within the data.

After the encoding phase, the extracted features are further refined using convolutional operations and max pooling, with Dropout applied to reduce the risk of overfitting. The final classification is performed via a linear transformation followed by a softmax layer, which maps the refined features to binary probabilities corresponding to either "fire" or "non-fire."

By integrating the global representation capabilities of the transformer with the local feature extraction strengths of convolutional operations, the proposed approach is particularly well-suited for classification tasks involving multi-sensor fire detection data.

For model training, the binary cross-entropy loss function is employed as the objective function, while the AdamW optimizer is selected, resulting in a model accuracy of 96.1%. The experimental results, presented in Fig. 6, demonstrate a steady decrease in the loss value on the validation set as the number of training iterations increases, with the model's accuracy gradually improving. Upon reaching 100 training cycles, the model's loss stabilizes, and the validation set accuracy achieves a peak value of 96.1%, indicating that the model has converged.

In this study, the Bert model and the Bert_C model were evaluated under identical experimental conditions. The results of these experiments are presented in Table 1. To assess the performance of the models in the time series classification task, four evaluation metrics were employed: precision, accuracy, recall, and F1 score. Greater values of these metrics signify superior classification performance.

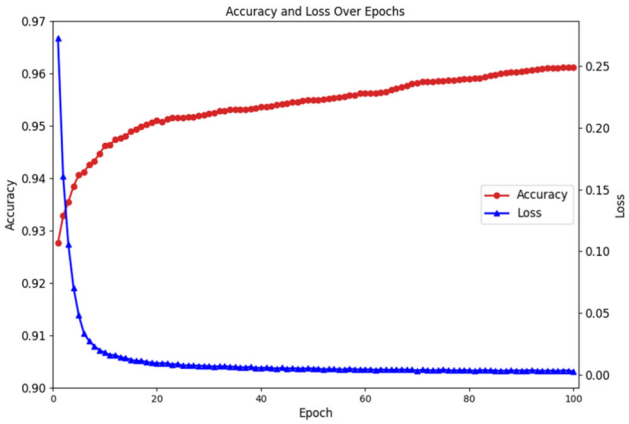


Fig.6 Training results of the Bert_C model

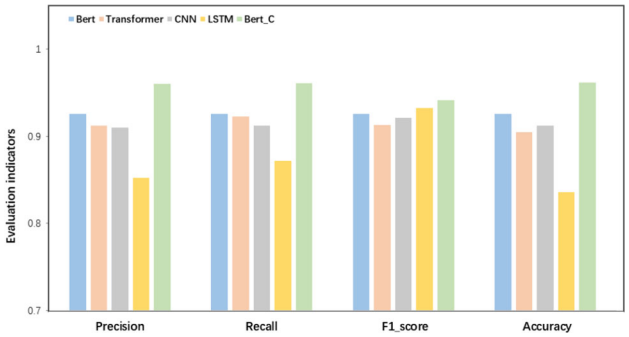


Fig.7 Evaluation results of different classification models

To further evaluate the performance of Bert_C, a classical deep learning model is selected, and a control group experiment is designed for a side-by-side comparison under identical training data. The experimental results are presented in Fig. 7.

Analysis of the experimental results reveals that the classification performance of the Bert model surpasses that of the traditional deep learning model. Furthermore, the Bert_C model demonstrates the optimal performance across all four evaluation metrics.

4.3 Digital twin

Finally, the equipment, data, and the developed fire detection model have been integrated into the smart laboratory fire safety management platform. The architecture of this platform is depicted in Fig. 8. It encompasses three core functional modules: real-time fire information, fire risk assessment, and fire facility control. The first module is designed to display changes in the concentrations of CO, CO2 detected by sensors during a fire. The sensor data serve not only as a critical input for the digital twin system but also directly feed into the deep learning model, enabling the

Table 1 Bert optimization experiments

Models	Evaluation indicators			
	Precision	Recall	F1_score	Accuracy
Bert	0.9257	0.9255	0.9256	0.9257
Bert_C	0.9605	0.9623	0.9615	0.9618



Fig.8 Digital twin platform for fire safety in smart labs



Fig.9 Fire test demonstration in a real scenario

platform to accurately predict and respond swiftly to fire situations. The second module is used to present the detected fire conditions and perform real-time assessments of the current fire status. The third module consolidates the control and management functions of various fire safety facilities within the laboratory. This includes functionalities such as the remote activation of the sprinkler system, all of which are easily accessible through the platform's intuitive interface. This design ensures that users can efficiently mobilize resources to minimize potential fire-related losses.

The system flows as follows. When a warning notification is triggered in the fire-monitoring system interface and an alarm is activated, it typically indicates the presence of a potential fire hazard. In this interactive interface, the left panel prominently displays the exact time and location of the incident, allowing users to access critical information immediately. A sprinkler control module is positioned in the lower-left corner of the interface. Users can monitor real-time surveillance footage to visually assess the fire scene, and subsequently make an informed decision regarding the activation of sprinkler systems based on actual conditions. The right side of the interface presents current meteorological information, including temperature, wind speed, and other relevant weather parameters. These data are essential for understanding the environmental context of the fire and for anticipating its potential spread.

Additionally, the lower section of the interface displays trend graphs of indoor carbon monoxide (CO) and carbon dioxide (CO_2) concentrations. These visualizations assist users in analyzing indoor air quality and serve as an important reference for formulating appropriate fire response strategies. This is demonstrated in Fig. 9.

Although the system successfully facilitates interaction between the model and the real physical environment, providing substantial support for fire prediction and emergency response, certain challenges remain in practical deployment and application. Notably, the reliability of sensor communication is a critical issue. The complex laboratory environment, coupled with the inherent uncertainty of fire scenarios, can result in interruptions or distortions in sensor data transmission. Additionally, cost considerations present a significant constraint for the digital twin framework. Highly accurate ground sensors are typically associated with high costs, making large-scale deployment financially challenging.

The digital twin framework proposed in this paper demonstrates a significant level of versatility and flexibility, making it suitable for the management of various types of laboratories. The core functionalities of the framework include data communication management, fire prediction modeling, and fire safety management. The modular design enhances the adaptability and extensibility of the framework, enabling its direct application to laboratory environments of different sizes and types.

5 Conclusions

This paper proposes a digital twin-based framework for intelligent building fire safety management. The framework, which integrates the Internet of Things, is demonstrated with

a real-time fire recognition model that utilizes deep learning algorithms to enable fire detection and monitoring. A fire-monitoring digital twin platform was developed within the Unity 3D environment, which facilitates human–computer interaction. The accuracy of the developed predictive model and the overall system capabilities were subsequently validated through numerical experiments. The results indicate that both the proposed framework and the predictive model are effective for simulating and managing full-scale building fire scenarios. The platform supports routine fire safety management and emergency response, with the system capable of rapidly detecting fires and triggering the alarm mechanism automatically.

The proposed digital twin-based fire protection system breaks through the limitation that the traditional system is triggered only when the fire characteristics are obvious; integrates multi-source sensor data to build a real-time 3D fire model; visualizes the development of the fire dynamics (including smoke diffusion, temperature distribution, etc.); solves the defects of the traditional system of a single alarm and the lack of spatial information; and integrates the Internet of Things (IoT), artificial intelligence and 3D visualization technologies; and builds a closed-loop "sensing-analysis-warning-disposal" system. It integrates the Internet of Things, artificial intelligence, and three-dimensional visualization technology, and builds a closed-loop "perception-analysis-warning-disposal" system.

The system is highly scalable and adaptable, and can be flexibly configured to fit the specific characteristics of various building types, including office buildings, shopping centers, and residential structures. By fine-tuning the threshold and weight parameters of the fire recognition algorithm, the system can be effectively adapted to different spatial topologies, such as large-span atriums or densely partitioned interior spaces. In addition, the system supports deep integration with structured data imported from building information modeling (BIM), which facilitates a more comprehensive understanding of the spatial and functional layout of a building.

In addition, the system is capable of incorporating a wide range of fire protection equipment, such as gas-based suppression systems and smoke extraction devices. Tailored emergency response plans can be developed based on the specific functional requirements and risk profiles of different building types. Through these capabilities, the system enables precise and efficient fire prevention and control across a broad spectrum of architectural environments.

However, several limitations exist in this study, including the absence of real-world data and the lack of realistic considerations within the proposed framework. Additionally, the evaluation of the sensor network and predictive model is insufficient, and the adaptability of the framework to various scenarios remains untested. These challenges can be addressed by collecting more comprehensive, high-quality data from real fire incidents, adopting advanced modeling techniques, and conducting extensive testing and validation.

Author Contribution T.L.T、S.F.S and C.L.Z conceived and designed the study, and wrote the main manuscript.T.L.T was responsible for review, editing, and supervision. C.L.Z performed the experimental operations and data processing. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of Interests The authors declare no competing interests.

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